

ch 25: Capacitance

zero potential energy

A diagram showing two positive charges, represented by circles with a '+' sign inside, connected by a horizontal dashed line. Below the line, the letter 'r' indicates the distance between the two charges.

(scalar, no direction)

ex 2):

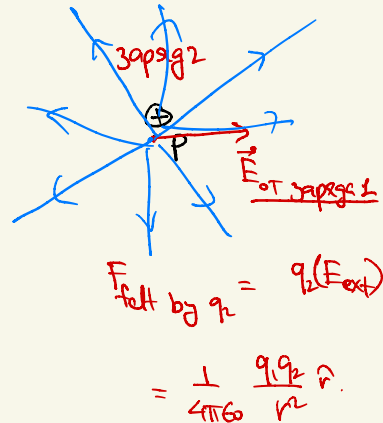
The diagram illustrates the work done by an electric field. A positive charge (+) is moved from a distance r_0 to a distance r from a fixed positive charge (+). The work done by the field is labeled as $PE = - W_{done \text{ by field}}$.

PE = - W_{done} by field

A diagram showing a positive point charge, represented by a circle with a '+' sign. Five black lines with arrows radiate outwards from the charge, representing electric field lines. One line extends horizontally to the left. A blue dot is placed on this horizontal line to the left of the charge. A blue arrow points from the charge towards the blue dot. To the left of the charge, the text $E_L = \frac{kq_L}{r_L^2}$ is written in blue. Above the charge, the word 'заряд' (charge) is written in red.

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$F_{\text{ext}} \text{ by } q_1 = q_1 F_{\text{ext}} \\ = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$



$$\text{Fetto by } g = 0$$

Let's go back to view 2



$$= - \int_{\infty}^r \frac{k q_1 q_2}{r^2} \cdot dr$$

$$PE = \frac{kg_1 r_1}{r}$$

Негативная жертва: притеснение
 позитивная жертва: отказывать

Electric Potential:

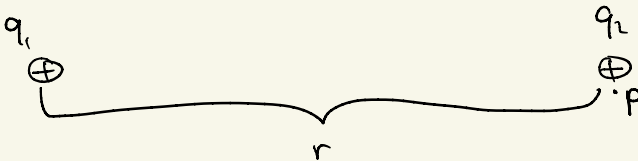
$$\frac{PE}{\text{unit charge}}$$

$$\vec{E} = \frac{kq}{r^2} \hat{r} \quad (\text{vector})$$

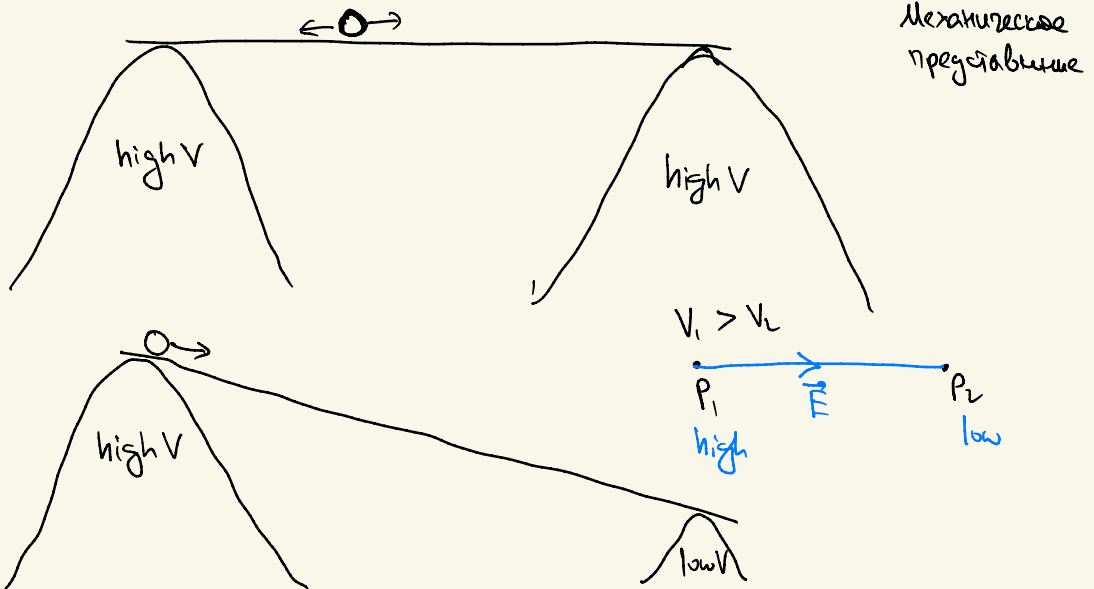
$$V = \frac{kq}{r} \quad (\text{scalar})$$

$q \oplus$

$$V = \frac{PE}{q}$$

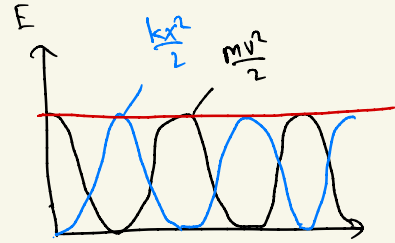
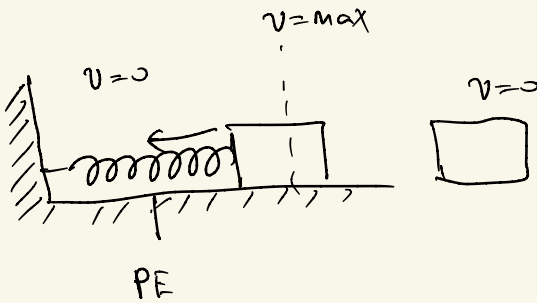


$$PE = \frac{kq_1q_2}{r}, \quad V(p) = \frac{kq_1}{r}$$

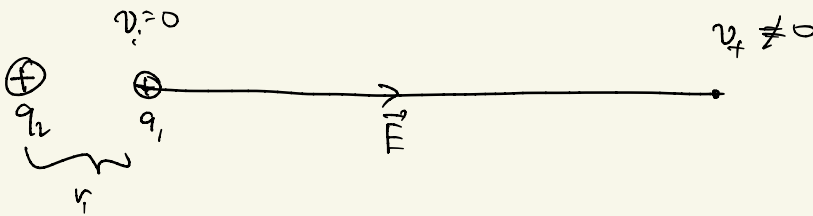


Isolated system:

$$E_{\text{mech}} = KE + PE$$



$$PE = PE_{\text{mech}} + PE_{\text{electrical}}$$



$$KE_i + PE_i = KE_f + PE_f$$

$$\frac{mv^2}{2} + \frac{kq_1q_2}{r_1} + \frac{GM_1M_2}{r} = \frac{mv^2}{2} + \frac{kq_1q_2}{r_2} + \frac{GM_1M_2}{r_2}$$

we can ignore negligible
no problems < energy

$$\frac{F_{\text{em}}}{F_g} = \frac{G}{k} 10^4$$

$$\Delta PE = KE$$

$$G = 6.67 \times 10^{-11}$$



$$k = 9 \times 10^9$$

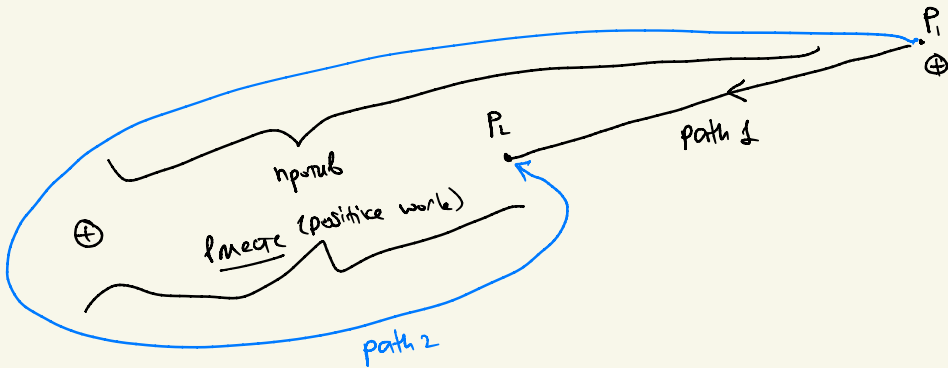
$$PE = -W_{\text{done by field}}$$

$$U = PE = - \int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{r} \rightarrow \text{messy}$$

$$\star - \frac{dU}{dr} = F$$

this Itho (2015)
18.1, 22

- Conservative fields: path independent

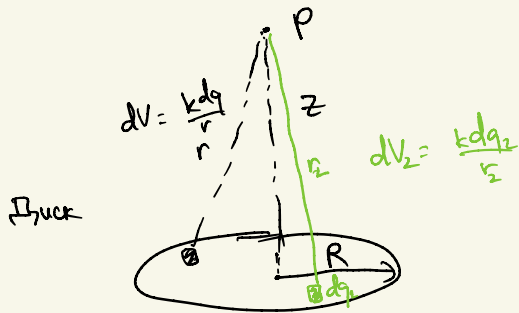


W_{done by field, path 1} $\stackrel{?}{=}$ W_{done by field, path 2}

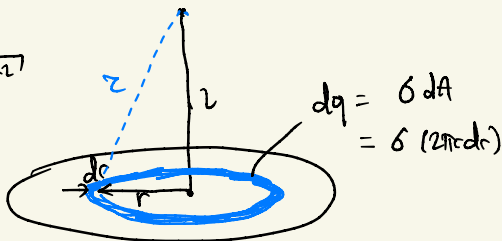
$$W_{\text{done by } w} = -\Delta PE$$

$$= -(PE_f - PE_i)$$

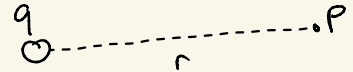
Ex: find $V(P) = ?$



$$\sigma = \frac{Q}{\pi R^2}$$



тогорино згарз



$$V(r) = \frac{kq}{r}$$

$$V = \sum_{i=1}^N dV_i$$

$$= \sum_{i=1}^N \frac{k dq_i}{r_i}$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

$$V = \int dV = \int \frac{k dq}{r} = k \int_0^R \frac{\sigma \cdot 2\pi r dr}{\sqrt{r^2 + z^2}}$$

$$= k 2\pi \int_0^R \frac{r dr}{\sqrt{r^2 + z^2}}$$

$$V = k 2\sigma\pi \int_0^R \frac{\cancel{rd}r}{\sqrt{r^2+z^2}}$$

$$E_{\text{field}} = k\sigma 2\pi \int_0^R \frac{rd r}{(r^2+z^2)^{3/2}}$$

Method of Substitution:

$$\text{let } u = r^2 + z^2$$

$$\frac{du}{dr} = 2r \Rightarrow r dr = \frac{du}{2}$$

$$V = 2k\sigma\pi \int_{u_i}^{u_f} \frac{du/2}{\sqrt{u}} = \frac{2k\sigma\pi}{2} \int u^{-1/2} du$$

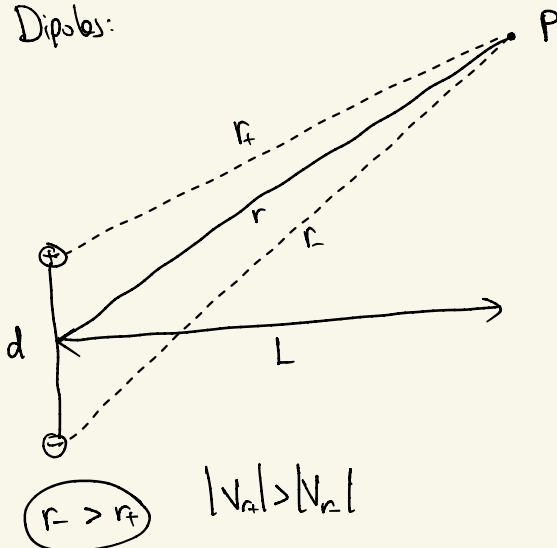
$$= k\sigma\pi \frac{\sqrt{u}}{1/2} \Big|_{u_i}^{u_f}$$

$$= 2k\sigma\pi \sqrt{r^2+z^2} \Big|_0^R$$

$$V(P) = 2k\sigma\pi (\sqrt{R^2+z^2} - \sqrt{z^2})$$

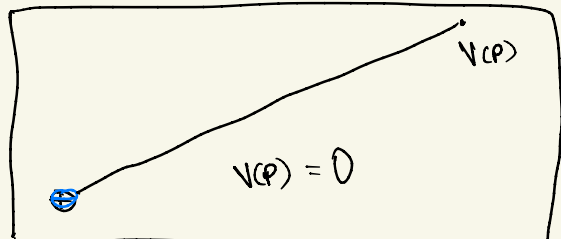
Примеры кондензаторов: сферическое плоскостное газ вакуум

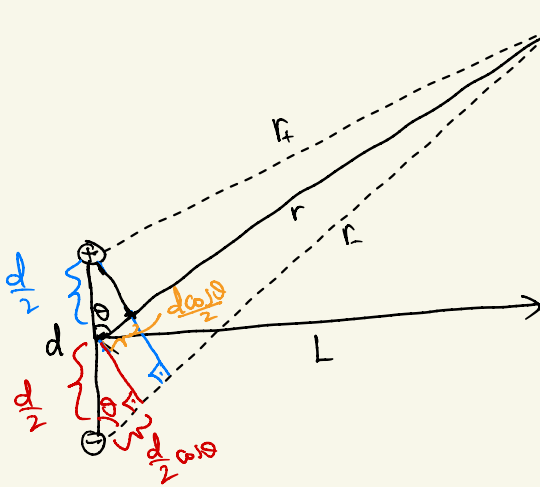
Dipoles:



$$L \gg d$$

From long distance, it seems $\oplus, \ominus$ merge together, but there's some quadratic & higher order terms





$$V(P) = V_{r_+} + V_{r_-}$$

$$V_{r_+} = \frac{kq}{r_+}$$

$$V_{r_-} = \frac{k(-q)}{r_-}$$

$$r_+ \approx r - \frac{d \cos \theta}{2}$$

$$r_- \approx r + \frac{d \cos \theta}{2}$$

$$V(P) = V_{r_+} + V_{r_-}$$

$$= \frac{kq}{r_+} - \frac{kq}{r_-}$$

$$= kq \left[\frac{1}{\left(r - \frac{d \cos \theta}{2}\right)} - \frac{1}{\left(r + \frac{d \cos \theta}{2}\right)} \right]$$

In these kind of situation : Binomial Expansion

$$(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$+ \frac{n(n-1)\dots(n-r+1)}{r!}x^r + \dots \text{terms up to } \infty$$

कोгда y так : $x \ll 1$

$$(1+x)^n \approx 1 + nx$$

$$V = kq \left[\frac{1}{r \left(1 - \frac{d \cos \theta}{2r}\right)} - \frac{1}{r \left(1 + \frac{d \cos \theta}{2r}\right)} \right]$$

$$\left| \begin{array}{l} d \ll r \\ 1 \ll \frac{d}{r} \end{array} \right.$$

$$V = kq \left[\frac{1}{r \left(1 - \frac{d \cos \theta}{2r}\right)} - \frac{1}{r \left(1 + \frac{d \cos \theta}{2r}\right)} \right] \quad \left. \vphantom{\frac{1}{r \left(1 - \frac{d \cos \theta}{2r}\right)}} \right\} \frac{1}{a} = a^{-1}$$

$$= \frac{kq}{r} \left[\left(1 - \frac{d \cos \theta}{2r}\right)^{-1} - \left(1 + \frac{d \cos \theta}{2r}\right)^{-1} \right]$$

$$= \frac{kq}{r} \left[1 + \frac{d \cos \theta}{2r} - \left(1 - \frac{d \cos \theta}{2r}\right) \right]$$

$$= \frac{kq}{r} \left[\cancel{2} \frac{d \cos \theta}{\cancel{2}r} \right]$$

$$= \frac{kqd}{r^2} \cos \theta$$

$$\boxed{V_p = \frac{kp \cos \theta}{r^2}}$$



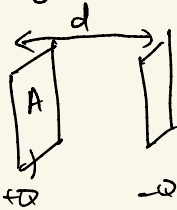
$$\boxed{\vec{p} = qd}$$

Chapter 25: Capacitance

Capacitance (ёмкость) - is the ability of a system (geometry) to store electric charge (соответственно заряду).

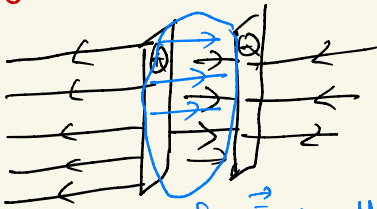
Battery

Most commonly known: parallel-plate capacitors



$$C = \frac{\epsilon_0 A}{d}$$

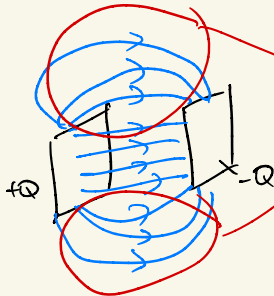
Ideally ($A \rightarrow \infty$): uniform fields



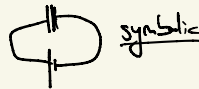
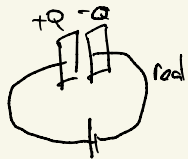
Uniform $\vec{E}$ in the approximation

where $A \rightarrow \infty$ идеально
бесконечная
пластина

But in reality



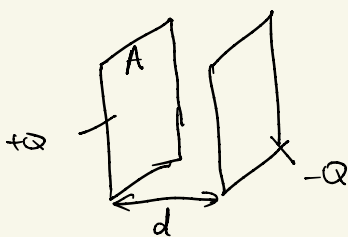
Edge effects
(краевые эффекты)



Главное уравнение ёмкости:

$$C = \frac{Q}{\Delta V}$$

Here, you take Q that has moved from one plate to another

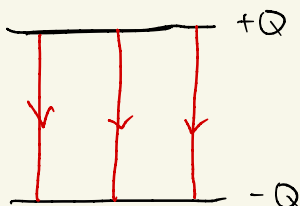


$$C = \frac{Q}{\Delta V}$$

$$\Delta V = - \int \vec{E} \cdot d\vec{r}$$

$$\Delta PE = - \underbrace{\int \vec{F} \cdot d\vec{r}}_W$$

$$E = 0$$



$$E = 0$$

$$\vec{E} = \text{const} = \frac{\sigma}{\epsilon_0}$$

$$\vec{E} = \frac{\Delta V}{d}$$

$$\Delta V = (E)d = Ed$$

$$\text{if } E = \text{const.}$$

in general,

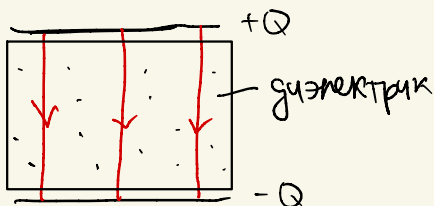
$$\Delta V = \int \vec{E} \cdot d\vec{r}$$

$$\vec{E} = - \frac{dV}{dr}$$

$$\Delta V = - \int \vec{E}(r) \cdot d\vec{r}$$

A capacitor consists of two conductors

$$E = 0$$



$$E = 0$$

$$\vec{E} = \text{const} = \frac{\sigma}{\epsilon}$$

2 steps to find C of the system

$$\Delta V = E \cdot d$$

step 1

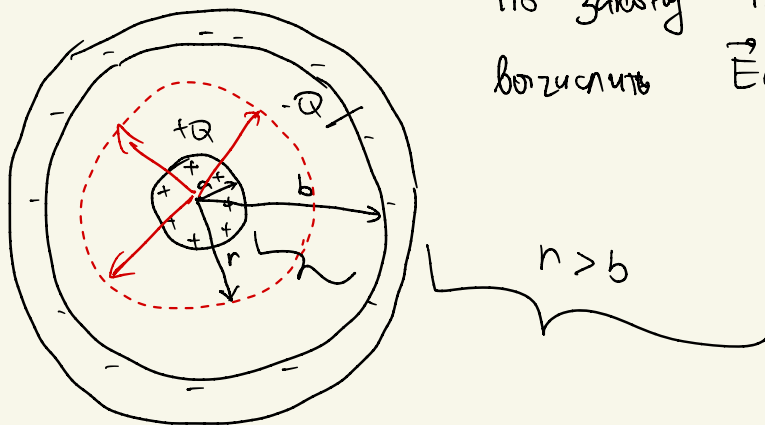
$$C = \frac{Q}{\Delta V}$$

step 2

$$= \frac{Q}{Ed} = \frac{Q}{\frac{\sigma d}{\epsilon_0}} = \frac{Q}{\frac{Q}{A} \frac{d}{\epsilon_0}}$$

$$= \epsilon_0 \frac{A}{d}$$

$$\epsilon \frac{A}{d} \text{ (нм густота)}$$



По закону Гаусса, мы можем
вычислить $\vec{E}(r)$ ($r > b$)
($a < r < b$)
($r < a$)

$r > b$

• $E(r > b) = 0$

$$q_{in} = +Q - Q = 0.$$

$$\oint \vec{E} \cdot d\vec{a} = \frac{q_{in}}{\epsilon_0} \rightarrow E(r > b) = 0.$$

• $E(a < r < b) =$

$$\oint \vec{E} \cdot d\vec{a} = \frac{q_{in}}{\epsilon_0} = \frac{+Q}{\epsilon_0}$$

$$\downarrow$$

$$E(4\pi r^2) = \frac{+Q}{\epsilon_0}$$

$$\boxed{\vec{E}(a < r < b) = \frac{Q}{4\pi\epsilon_0 r^2} (\hat{r})}$$

• $E(r < a) = 0$

bec $q_{in} = 0$ again.

Remember, to compute C we needed 2 steps:

- step 1: ΔV
- step 2: $C = \frac{Q}{\Delta V}$

Step 1:

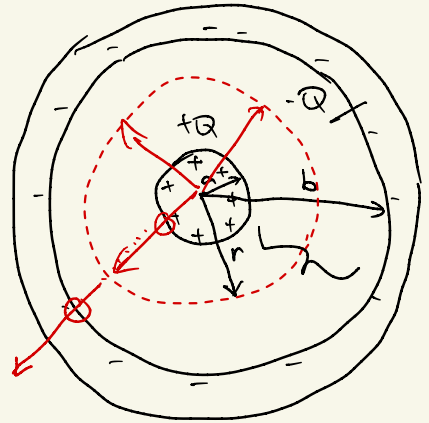
ΔV :

$$\vec{E}(r) = -\frac{dV}{dr} \rightarrow$$

$$\Delta V = - \int_{\vec{r}}^{\vec{r}_1 \vec{r}_2} \vec{E} \cdot d\vec{r}$$

$$\begin{aligned} \Delta V &= - \int_b^a \vec{E}(r) \cdot d\vec{r} \\ &= - \int_b^a \frac{+Q}{4\pi\epsilon_0 r^2} dr \\ &= - \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{r} \right) \Big|_b^a \end{aligned}$$

$$\Delta V = - \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)$$



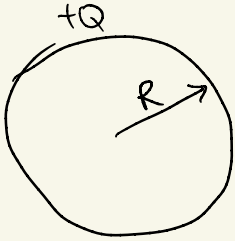
Step 2:

$$C = \frac{Q}{\Delta V}$$

$$C = \frac{Q}{\frac{Q}{4\pi\epsilon_0} \frac{(b-a)}{ab}} = \frac{4\pi\epsilon_0 ab}{(b-a)}$$

Capacitance of spherical concentric shell combination

Case 3: Capacitance of an isolated sphere



Геометрия: бесконечно большое пространство (в центре проводящая сфера с зарядом $+Q$).

Hint: remember 2 steps:

1) ΔV

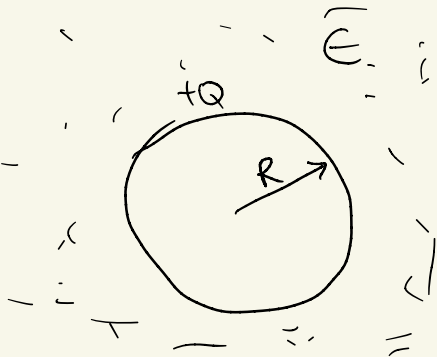
2) $C = \frac{Q}{\Delta V}$

$$|\Delta V| = |V_{\infty} - V_R|$$

$$= \left| 0 - \frac{Q}{4\pi\epsilon_0 R} \right|$$

$$= \frac{Q}{4\pi\epsilon_0 R}$$

$$C = \frac{Q}{\frac{Q}{4\pi\epsilon_0 R}} = 4\pi\epsilon_0 R$$

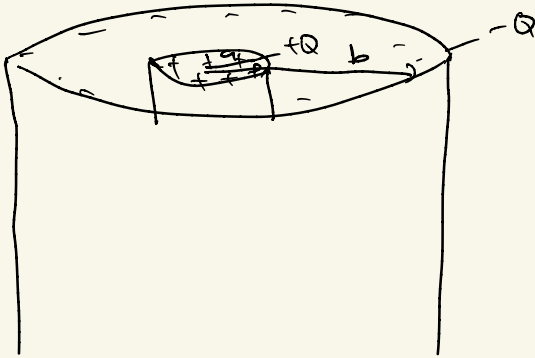


$$C = 4\pi\epsilon_0 R$$

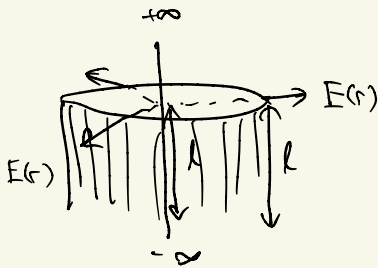
Case 4: Cylindrical geometry:

Haupta C = ?

$$\Delta V = - \int_a^b \vec{E} \cdot d\vec{r}$$



$$\vec{E}(r) \begin{cases} E = 0 & (r > b) \\ E = \frac{\lambda}{2\pi\epsilon_0 r} & (a < r < b) \\ E = 0 & (r < a) \end{cases}$$



$$\oint \vec{E} \cdot d\vec{a} = \frac{q_{in}}{\epsilon_0} = \frac{\lambda l}{\epsilon_0}$$

$$E(2\pi r l) = \frac{\lambda l}{\epsilon_0} \quad \lambda = \frac{Q}{l}$$

$$\boxed{\vec{E}(r) = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}}$$

$$\sigma = \frac{Q}{A_{area}} = \frac{Q}{2\pi a \cdot l}$$

$$\lambda = \frac{Q}{l} \rightarrow \boxed{\lambda = \sigma \cdot 2\pi a}$$

Step 1:

$$\Delta V = - \int_b^a \vec{E}(r) \cdot d\vec{r}$$

$$= - \int_b^a \frac{\lambda}{2\pi\epsilon_0 r} \cdot dr$$

$$= - \frac{\lambda}{2\pi\epsilon_0} \ln r \Big|_b^a$$

$$= - \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{a}{b}\right)$$

$$= \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

$$\ln a - \ln b = \ln\left(\frac{a}{b}\right)$$

$$\ln b - \ln a = \ln\left(\frac{b}{a}\right)$$

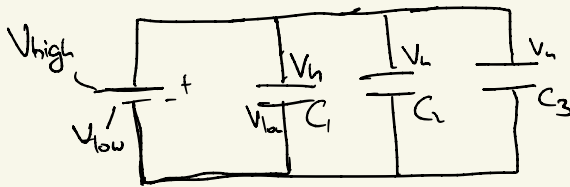
Step 2:

$$C = \frac{Q}{\Delta V}$$

$$= \frac{\cancel{\lambda} l}{\cancel{\lambda} \ln\left(\frac{b}{a}\right) 2\pi\epsilon_0}$$

$$C = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)}$$

Capacitors connected in parallel : $C_{eq} = ?$



$$\Delta V = V_{high} - V_{low}$$

ΔV gnx box $C \rightarrow$ ogunharobes

$$Q_{tot.} = C_{eq} V \rightarrow \Delta V = \frac{Q_{tot.}}{C_{eq.}}$$

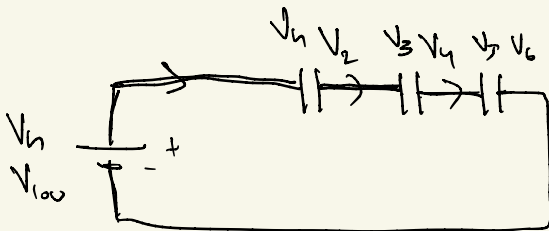
$$Q_{tot.} = q_1 + q_2 + q_3 = C_1 V + C_2 V + C_3 V$$

$$\frac{Q_{tot.}}{V} = C_{eq.} = C_1 + C_2 + C_3$$

$$V_{tot.} = V_1 = V_2 = V_3$$

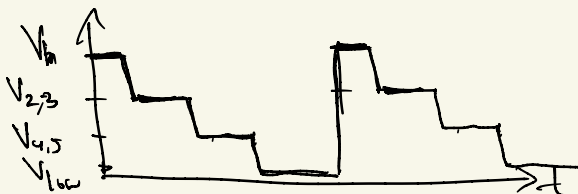
$$Q_{tot.} = Q_1 + Q_2 + Q_3$$

Capacitors connected in series : $C_{eq} = ?$



$$V_1 \neq V_2 \neq V_3$$

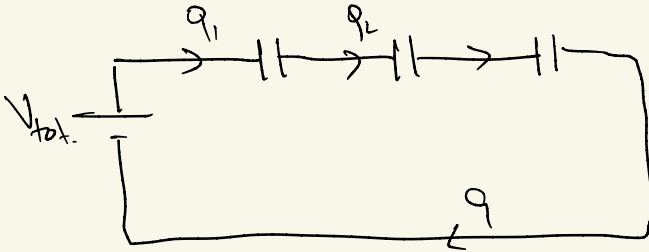
Voltage drop



$$V_{total} = V_1 + V_2 + V_3$$

$$V_{\text{total}} = V_1 + V_2 + V_3$$

$$V_{\text{tot.}} = \frac{Q_{\text{tot.}}}{C_{\text{tot.}}} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \frac{Q_3}{C_3}$$



$$q_1 = q_2 = q_3$$

same current is flowing across all capacitors

$$V_{\text{tot}} = \frac{\cancel{Q}}{C_{\text{eq}}} = \frac{\cancel{Q}}{C_1} + \frac{\cancel{Q}}{C_2} + \frac{\cancel{Q}}{C_3}$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Farad = unit of capacitance

$$C_1 = 20 \mu\text{F}, \quad C_2 = 20 \mu\text{F}$$

Case 1: Parallel connection

$$C_{\text{eq}} = C_1 + C_2 = 40 \mu\text{F}$$

результат

Case 2: Series connection

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} \rightarrow C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$$

$$= 10 \mu\text{F}$$

результат

Одни из главных skills (навыков, умений) - это умение работать с разными геометриями.

Two steps

Step 1:

Найти

$$\Delta V = - \int \vec{E}(r) \cdot d\vec{r}$$

Step 2:

$$C = \frac{Q}{\Delta V}$$

HW Ch 24

Checkpoint 3, 24-10 just read carefully

$$\Delta V = - \int \vec{E} \cdot d\vec{r}$$

Problems: 10, 11, 17, 23, 26, 31, 59, 72, 74

HW Ch 25

Problems: 9, 15, 18, 20, 21, 22, 23, 48, 49, 50, 59, 75

Deadline: Thu Aug 14th noon 12 PM.

